LOCAL RINGS UNDER A GROUND FIELD EXTENSION: FAITHFUL FLATNESS AND THE RESIDUE FIELD AT A POINT OF THE BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the parenthetical '(details omitted)' at the identification of the prime used for the localization; the parenthetical is reproduced verbatim. The surrounding factorization and the description of the prime are written out, so the omitted check is bounded.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-change-fields-flat is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- A local flat homomorphism of local rings with a residue field extension and a nilpotent maximal ideal structure is faithfully flat. Algebra, Lemma algebra-lemma-local-flat-ff, invoked at proof lines 258-259. Supplies faithful flatness in part (1).
- Points of a fibre product and their residue fields. Schemes, Lemma schemes-lemma-points-fibre-product, invoked at proof lines 260-261. Supplies the description of the residue field at a point of the base change used in part (2).
- Exactness of localization and of tensoring with a field extension. Standard, used at proof lines 279-280. Supplies part (4) from part (3).

Source reference. Original lemma 00HR: sources/stacks-chow/tags.txt, source lines 653–653. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01JT: sources/stacks-chow/tags.txt, source lines 2026–2026. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let K/k be an extension of fields. Let X be scheme over k and set $Y = X_K$. If $y \in Y$ with image $x \in X$, then*

(1) *$\mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$ is a faithfully flat local ring homomorphism,*
(2) *with $\mathfrak{p}_0 = \mathrm{Ker}(\kappa(x) \otimes_k K \to \kappa(y))$ we have $\kappa(y) = \kappa(\mathfrak{p}_0)$,*

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(3) $\mathcal{O}_{Y,y} = (\mathcal{O}_{X,x} \otimes_k K)_{\mathfrak{p}}$ *where* $\mathfrak{p} \subset \mathcal{O}_{X,x} \otimes_k K$ *is the inverse image of* $\mathfrak{p}_0$.
(4) *we have* $\mathcal{O}_{Y,y}/\mathfrak{m}_x\mathcal{O}_{Y,y} = (\kappa(x) \otimes_k K)_{\mathfrak{p}_0}$

Proof. We may assume $X = \mathrm{Spec}(A)$ is affine. Then $Y = \mathrm{Spec}(A \otimes_k K)$. Since K is flat over k, we see that $A \to A \otimes_k K$ is flat. Hence $Y \to X$ is flat and we get the first statement if we also use Algebra, Lemma 00HR. The second statement follows from Schemes, Lemma 01JT. Now y corresponds to a prime ideal $\mathfrak{q} \subset A \otimes_k K$ and x to $\mathfrak{r} = A \cap \mathfrak{q}$. Then $\mathfrak{p}_0$ is the kernel of the induced map $\kappa(\mathfrak{r}) \otimes_k K \to \kappa(\mathfrak{q})$. The map on local rings is

$$A_{\mathfrak{r}} \longrightarrow (A \otimes_k K)_{\mathfrak{q}}$$

We can factor this map through $A_{\mathfrak{r}} \otimes_k K = (A \otimes_k K)_{\mathfrak{r}}$ to get

$$A_{\mathfrak{r}} \longrightarrow A_{\mathfrak{r}} \otimes_k K \longrightarrow (A \otimes_k K)_{\mathfrak{q}}$$

and then the second arrow is a localization at some prime. This prime ideal is the inverse image of $\mathfrak{p}_0$ (details omitted) and this proves (3). To see (4) use (3) and that localization and $- \otimes_k K$ are exact functors. □